

Research Brief: Criticism Narratives and US Equity-Market Performance

Collaborative MVP Plan

June 1, 2026

1 Project Aim

This project studies whether criticism narratives about the US stock market are related to later market behaviour.

The main research question is:

Can criticism of the US equity market, extracted from public text data, explain or predict future returns, volatility, drawdowns, or reversals after controlling for recent market conditions?

The project combines two components:

1. **Discourse/annotation component:** defining, identifying, and validating what counts as market criticism.
2. **Quantitative/data component:** constructing criticism indices and testing their relationship with market outcomes.

2 Team Roles

Area	Angze: Quant/Data Lead	K: Discourse/Annotation Lead
Research design	Market variables, models, controls	Criticism definition, framing categories
Data collection	APIs, cleaning, deduplication	Source-quality review, sample inspection
Annotation	Export samples, compute validation metrics	Build codebook, label examples, resolve ambiguity
NLP	Feature extraction, classifiers, robustness	Evaluate whether labels capture real criticism
Market analysis	Returns, volatility, regressions, event studies	Interpret narrative mechanisms
Writing	Data, methods, results, figures	Definition, coding rules, interpretation, limitations

The MVP should be built so that K's contribution is not merely manual labelling. K should define the conceptual boundaries of criticism and help validate whether the automated extraction is meaningful.

3 Operational Definition of Criticism

A text item counts as market criticism only if it satisfies both conditions.

1. Market Target Condition

The item must refer to the US equity market or a major US equity benchmark, such as:

US stock market, Wall Street, S&P 500, Nasdaq, US equities, American stocks, Magnificent Seven, AI stocks, tech stocks.

2. Critical Stance Condition

The item must express concern, scepticism, warning, or negative evaluation of the market’s valuation, structure, sustainability, speculation, concentration, or fragility.

For the MVP, use five criticism categories:

1. **Valuation criticism:** the market is too expensive or detached from fundamentals.
2. **Bubble/speculation criticism:** the market is described as a bubble, mania, frenzy, or casino-like environment.
3. **Crash/correction warning:** the text warns of a crash, selloff, correction, or collapse.
4. **AI/tech-hype criticism:** AI or technology stocks are described as overhyped or unsustainable.
5. **Concentration criticism:** the market is criticised for relying too heavily on a small number of mega-cap stocks.

K’s MVP task is to turn these categories into a clear annotation codebook with examples, boundary cases, and exclusion rules.

4 Exclusion Rules

A text item should not be counted as criticism if it is:

1. purely descriptive, e.g. “S&P 500 falls 1%”;
2. only about a single company with no wider market implication;
3. about non-US markets only;
4. a historical explainer not referring to current market conditions;
5. using negative keywords without a critical stance, e.g. “Bubble fears ease”;
6. a technical trading phrase rather than a criticism narrative.

K should lead the review of ambiguous cases and maintain a short decision log explaining how difficult examples are handled.

5 Data Sources

The MVP should use structured data sources before attempting broad web scraping.

Data Type	Suggested Source	Purpose
News headlines and metadata	GDELT DOC 2.0	Main text dataset
News robustness check	NewsAPI	Optional source comparison
Market prices	SPY, S&P 500, QQQ, RSP	Return and drawdown analysis
Volatility	VIX, realised volatility	Volatility channel
Macro controls	FRED / public macro data	Rates, inflation, event controls
Retail discourse, optional	Reddit API	Later extension beyond MVP

For the MVP, headline-level analysis is sufficient. Full article-body extraction can be added later if legally and technically feasible.

K’s role in the data-source stage is to review whether the selected sources represent different kinds of market discourse, such as institutional financial media, mainstream business news, and retail-oriented discussion.

6 Time Interval

Recommended MVP interval:

2022-01-01 to 2026-05-31

Expanded interval:

2020-01-01 to 2026-05-31

Daily frequency should be used. If article timestamps are available, articles published after 16:00 New York time should be assigned to the next trading day.

7 Extraction and Annotation Workflow

The MVP should use a hybrid workflow:

1. collect all US-market-related headlines;
2. collect candidate criticism headlines using predefined keyword queries;
3. deduplicate and clean the text data;
4. export a mixed annotation sample for K;
5. K labels market relevance, criticism presence, criticism category, and intensity;
6. use K’s labels to validate and refine the automated extraction;
7. construct daily criticism indices;
8. merge the indices with market data;
9. run market-performance tests.

Market Query

“US stock market” OR “Wall Street” OR “S&P 500” OR “Nasdaq” OR “US equities”
OR “American stocks”

Criticism Query

overvalued OR expensive OR bubble OR mania OR crash OR correction OR selloff
OR fragile OR frothy OR speculative OR “concentration risk” OR “AI bubble” OR
“Magnificent Seven”

The candidate criticism dataset is generated by:

$$Q_{\text{market}} \cap Q_{\text{criticism}}$$

8 K’s Annotation Deliverables

K should produce four MVP deliverables.

1. Annotation Codebook

The codebook should define:

1. what counts as market criticism;
2. each criticism category;
3. examples of included cases;
4. examples of excluded cases;
5. rules for ambiguous headlines;
6. intensity scale, if used.

2. Labelled Validation Set

K should label at least 500 headlines for the MVP. A preferred target is 1,000 headlines.

Each labelled item should include:

1. market relevance: yes/no;
2. criticism: yes/no;
3. criticism category;
4. criticism intensity, e.g. 0–3;
5. optional notes for ambiguous cases.

3. Ambiguity Log

K should maintain a short log of difficult examples, such as:

1. headlines containing criticism keywords but no critical stance;
2. neutral headlines with implied criticism;
3. single-stock criticism with possible market-wide relevance;
4. headlines mixing positive market performance with negative framing.

4. Interpretation Notes

K should write brief notes on what each criticism category may represent narratively. These notes will support the discussion section but do not need to become a full theory chapter for the MVP.

9 Minimal Samples

Item	Minimum	Preferred
Time span	2 years	4–6 years
Trading days	500	1,000+
Market-related headlines	10,000	30,000+
Candidate criticism headlines	2,000	10,000+
K-labelled headlines	500	1,000+
Criticism categories	3–5	5–8

For the MVP, precision is more important than recall. It is better to construct a smaller but cleaner criticism index than a large noisy one.

10 Criticism Index Construction

The main daily Market Criticism Index is:

$$MCI_t = \frac{\text{Number of criticism articles on day } t}{\text{Number of all US-market-related articles on day } t}$$

A rolling standardised version is:

$$MCI_t^z = \frac{MCI_t - \mu_{t,60}}{\sigma_{t,60}}$$

where $\mu_{t,60}$ and $\sigma_{t,60}$ are the 60-trading-day rolling mean and standard deviation.

Category-specific indices are:

$$MCI_{k,t} = \frac{\text{Number of criticism articles in category } k \text{ on day } t}{\text{Number of all US-market-related articles on day } t}$$

K's labels should be used to evaluate whether the automated index is capturing genuine criticism rather than generic negativity.

11 NLP Component

NLP should be used as a supporting tool, not as the sole definition of criticism.

Recommended NLP features:

1. VADER sentiment score;
2. TextBlob polarity and subjectivity;
3. financial-negative word counts;
4. uncertainty-word counts;
5. category keyword flags;
6. headline length;
7. punctuation/emphasis features;
8. optional supervised classifier trained on K's labels.

The preferred sequence is:

K's codebook → manual labels → rule-based extraction → NLP features → validation or classifier → Market Crit

TextBlob and VADER may be included as robustness features, but they should not define criticism. PCA is optional but may reduce interpretability. LASSO can be useful later for feature selection. ARIMAX may be useful for volatility, VIX, or volume, but simple regressions and event studies should remain the MVP baseline.

12 Market Index and Outcome Variables

The main market benchmark should be SPY or the S&P 500.

Secondary benchmarks:

1. QQQ or Nasdaq-100;
2. RSP or equal-weight S&P 500;
3. VIX;
4. SPY/QQQ trading volume.

Main outcomes:

1. next-day return;
2. next-week return;
3. next-month return;
4. future realised volatility;
5. future maximum drawdown;
6. VIX change.

Use log returns:

$$r_{t,h} = \log(P_{t+h}) - \log(P_t)$$

Use realised volatility:

$$RV_{t,h} = \sqrt{\sum_{i=1}^h r_{t+i}^2}$$

Use forward maximum drawdown:

$$DD_{t,h} = \min_{1 \leq i \leq h} \left(\frac{P_{t+i}}{P_t} - 1 \right)$$

Test horizons:

$$h \in \{1, 5, 21\}$$

13 Confounders

The empirical models should control for:

1. lagged market returns;
2. lagged VIX;
3. lagged realised volatility;
4. recent market momentum;
5. total market-news volume;
6. CPI, FOMC, and NFP event dummies;
7. day-of-week effects;
8. source composition;
9. market regime, such as high-VIX versus low-VIX periods.

A baseline return model is:

$$r_{t,h} = \alpha + \beta MCI_t^z + \gamma X_t + \epsilon_t$$

A baseline volatility model is:

$$RV_{t,h} = \alpha + \beta MCI_t^z + \gamma X_t + \epsilon_t$$

where X_t contains control variables. For overlapping horizons, Newey-West standard errors should be used.

14 Possible Mechanisms

The MVP should test four mechanisms:

1. **Reactive narrative:** criticism rises after market weakness.
2. **Warning signal:** criticism precedes weaker returns or larger drawdowns.
3. **Contrarian signal:** criticism spikes precede rebounds.
4. **Volatility/attention signal:** criticism predicts volatility, VIX, or volume rather than return direction.

K's role is to help interpret what each mechanism means narratively. The quantitative analysis should test whether these mechanisms appear in the data.

15 Limitations

Main limitations:

1. the dataset may not represent all market criticism;
2. keyword queries may miss subtle criticism;
3. generic sentiment tools may misread financial language;
4. criticism may be reactive rather than predictive;
5. article timestamps may not perfectly align with market timing;
6. source authority differs across outlets;
7. multiple testing may produce false positives;
8. correlation does not imply causation.

K should contribute specifically to the limitations concerning discourse interpretation, ambiguity, framing, and source context.

16 MVP Definition

The MVP should answer:

Does a validated daily Market Criticism Index extracted from financial-media headlines relate to future US equity returns, volatility, or drawdowns?

MVP Steps

1. collect GDELT headlines from 2022-01-01 to 2026-05-31;
2. extract US-market-related headlines;
3. extract candidate criticism headlines;
4. K creates the annotation codebook;
5. K labels 500–1,000 headlines;
6. validate the automated extraction against K's labels;
7. construct raw, normalised, and standardised criticism indices;
8. merge the indices with SPY, QQQ, RSP, VIX, and controls;
9. test reactive, predictive, contrarian, and volatility mechanisms;
10. produce figures, regression tables, and a concise working paper.

MVP Success Criteria

The MVP is successful if it produces:

1. a reproducible headline dataset;
2. a clear criticism codebook;
3. a labelled validation set from K;
4. a validated Market Criticism Index;
5. category-specific criticism measures;
6. market-performance tests;
7. interpretable figures and tables;
8. an honest conclusion, whether positive, negative, or mixed.

17 Proposed MVP Division of Labour

MVP Task	Angze	K
Define research question	Co-lead	Co-lead
Define criticism categories	Support technical feasibility	Lead
Build annotation codebook	Support CSV structure	Lead
Collect headline data	Lead	Review source relevance
Clean and deduplicate data	Lead	Review ambiguous examples
Create annotation sample	Lead	Label and comment
Validate extraction quality	Compute metrics	Interpret errors
Build criticism index	Lead	Review category meaning
Run market tests	Lead	Support interpretation
Write methods/results	Lead	Support
Write definition/discussion	Support	Lead
Final paper editing	Co-lead	Co-lead